Dwala Global Resources project

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Abstract

This report describes the design, algorithm support and implementation of the first release of the BlastSmart mobile application, namely the Surface Blasting Design Calculator and the Fragmentation Calculator. The traditional method which is inefficient, manual and susceptible to human error, is still in use for many blasting operations in South Africa. BlastSmart makes these calculations automated to make more accurate decisions on site. The Surface Blasting Design Calculator determines surface blast geometry and blast parameters and a Fragmentation Calculator models mean fragment size based on the known empirical blast models, including Kuz-Ram. The architecture, mathematical models, and test scenarios of the App are assessed as well, showing that real-time calculations with the app can significantly cut down the number of design iterations and improve rock fragmentation efficiency.

1 Introduction & Background

Drilling and blasting are the two most important processes in modern mining and civil excavation operations for rock fragmentation. Fragmentation is very important because it will affect downstream operation such as loading, hauling, crushing and milling. Traditionally, blasting engineers have used handwritten notebooks, paper design guidelines and static spreadsheets on site to make estimates of the explosive load, burden and stemming lengths.

Dwala Global Resources has begun work on the **BlastSmart App** to make these paper-based approaches digital. This project aims to work on Phase 1 development which will provide two key calculation modules:

1. **Surface Blasting Design Calculator:** Automates the geometric and explosive-loading parameters per blasthole.

2. **Fragmentation Calculator:** Predicts rock fragment size distribution to prevent under-fragmentation (boulders) or over-fragmentation (excessive fines).

2 Literature Review & Technical Formulation

To establish a robust mathematical foundation for the software, established empirical blasting formulas were integrated. by

2.1 Surface Blasting Design Formulations

The application prompts the user for four basic parameters: Blasting Diameter (D or BD), Average In-Hole Density (ρ), Spacing-to-Burden Ratio (S/B), and Bench Height (BH). The calculation of the charge weight per hole (M_c) and the expected outputs rely on volumetric and physical relationships:

$$M_c = \rho \times \frac{\pi \times D^2}{4} \times CL \tag{1}$$

Where:

- M_c = Charge weight per hole (kg)
- ρ = Average In-Hole explosive density (g/cm^3 or kg/m^3)
- D = Blasthole/Blasting Diameter (mm converted to meters)
- CL = Charge length (m), where $CL = BH - SL$ (Bench Height minus Stemming Length)

Stemming length (SL) is typically calculated using an empirical multiplier of the burden (B), such as:

$$SL = k_s \times B \tag{2}$$

2.2 Fragmentation Modelling

To estimate the mean fragment size (x_{50}), the system uses the **Kuz-Ram empirical model**:

$$x_{50} = A \times K^{-0.8} \times Q^{1/6} \times \left(\frac{115}{E}\right)^{19/30} \tag{3}$$

Where:

- A = Rock factor
- K = Powder Factor (kg/m^3)
- Q = Mass of explosive per hole (kg)
- E = Relative Weight Strength of explosive (ANFO = 100)

3 System Design & Architecture

The system architecture of the Blast-smart application was designed to be lightweight and executed offline on the mobile device to ensure it can be accessed when the mobile device is in remote locations within the mine site.

3.1 Module 1: Surface Blasting Design Flow

The application checks whether $BD > 0, \rho > 0, BH > 0$ when they are inputted. The spacing to burden ratio is entered by the user and the program calculates the burden (B) and spacing (S) according to normal blasting practices. It outputs:

- Expected Stemming Length (SL)
- Volumetric parameters and charge distributions.

3.2 Module 2: Fragmentation Calculator Flow

Taking the outputs of the blast design (burden, stemming, powder factor), this module computes:

- The mean fragment size (d_{50}) based on the relative effective energy of the explosive (E).
- Dynamic adjustments if the powder factor changes.

4 Results & Discussion

To validate the application, a simulated blasting scenario was processed through both the manual calculation flow and the developed software.

Table 1: Validation Dataset and System Outputs

Parameter	**Input Value**	**Calculated Output**
Blasting Diameter (BD)	200 mm	–
In-Hole Density (ρ)	1.2 g/cm^3	–
Bench Height (BH)	12 m	–
Calculated Stemming Length (SL)	–	[Insert your output, e.g., 4.2 m]
Calculated Charge Mass (M_c)	–	[Insert your output, e.g., 294 kg]
Predicted Mean Fragment Size (x_{50})	–	[Insert your output, e.g., 25 cm]

4.1 Discussion

These formulas can be automated and critical errors are prevented. For example, if the conversion of BD (from 200 mm to 0.2 m) was done wrong, it can be a tenfold error in calculating the charge mass. BlastSmart ensures this by validating and standardizing units internally. Furthermore, the real-time feedback loop enables engineers to immediately modify the value of the stemming length (SL) and see the immediate effect of the change on fragment size (x_{50}) without having to manually re-calculate.

5 Conclusions & Future Work

Phase 1 of the Blast-smart App is a successful implementation of the idea that critical blasting calculators can be effectively automated and made mobile. The Surface Blasting and Fragmentation calculators offer quick, accurate and reliable calculations compatible with the standard empirical models used in the mining industry.

For future developments, efforts are focused on Phase 2, which will involve inclusion of underground blasting design templates and introduction of peak particle velocity (PPV) vibration attenuation modeling to protect surrounding civil infrastructure.

References

[1] Kuznetsov, V. M. (1973). *The mean diameter of the fragments formed by blasting rock*. Soviet Mining Science, 9(2), 144-148.

[2] Cunningham, C. V. B. (1983). *The Kuz-Ram model for prediction of fragmentation from blasting*. First International Symposium on Rock Fragmentation by Blasting, Luleå, Sweden.